

Hotel Reservation & Check-In System

CS 342 Project

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Introduction

The Hotel Reservation & Check-In System is a Java Swing-based desktop application developed for the CS342 course. The project aims to automate hotel front-desk operations by providing an efficient system for managing room reservations, guest check-in and check-out, billing, and payments.

The application follows the MVC architecture and integrates with a relational database using JDBC to ensure organized data management and persistence. It supports role-based authentication for receptionists and managers, along with input validation and exception handling to prevent booking conflicts and invalid operations.

This project demonstrates the practical use of GUI development, database integration, and software engineering principles in building a realistic hotel management system.



Application Overview

Hotel Reservation System is a desktop-based application designed to support the daily operations of a hotel in an organized and efficient manner. It manages the complete reservation lifecycle, beginning with guest registration and reservation creation, continuing through check-in and check-out processes, and concluding with billing and payment tracking.

The application enables hotel staff to create and manage guest records and reservations while ensuring accurate tracking of room availability and room statuses, including Available, Occupied, and Cleaning Required. This functionality provides real-time visibility into room usage and operational readiness.

During the check-in and check-out processes, the system enforces strict status validation to ensure that reservations and rooms follow the correct operational workflow. This helps prevent conflicts such as double bookings and invalid room assignments.

At check-out, the system generates detailed and itemized invoices that include room charges, additional services, taxes, and applicable discounts. Discounts are retrieved from the database and applied according to predefined rules, allowing the system to accurately calculate the final payable amount.

The billing module supports partial and multiple payments by allowing more than one payment record to be associated with a single invoice. This enables flexible payment handling while maintaining accurate financial records.

In addition, the system provides reporting features such as income summaries, enabling management to analyze financial performance by comparing total invoiced amounts with received payments, including the impact of applied discounts. By integrating all system components with a centralized database, the application ensures data accuracy, consistency, and efficient hotel operations.

GUI Elements

The interface is designed to be simple, intuitive, and easy to navigate, allowing hotel staff to perform daily operations efficiently.

The system includes multiple screens such as Login, Manager Dashboard, Reservation Management, Room Management, Check-In, Check-Out, Invoice View, Payment Management, and Discount Management. Each screen is organized using standard Swing components including buttons, text fields, labels, combo boxes, tables, and dialog windows.

Tables (JTable) are used extensively to display structured data such as reservations, rooms, invoices, payments, and discounts. These tables support row selection, enabling users to view, update, or delete records easily.

Dialog windows (JDialog) are used for focused tasks such as check-in, check-out, invoice viewing, and payment entry, ensuring that users complete one operation at a time without confusion.

Buttons are clearly labeled and styled consistently across the application to guide users through actions such as creating records, confirming operations, generating invoices, and completing payments. Confirmation and warning dialogs are displayed for critical actions to prevent accidental data loss.

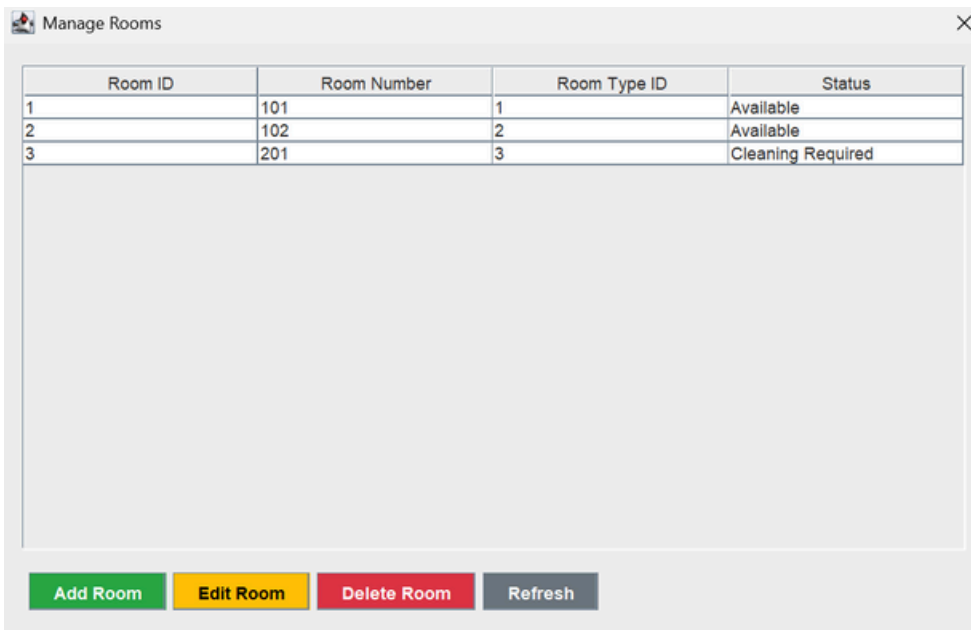
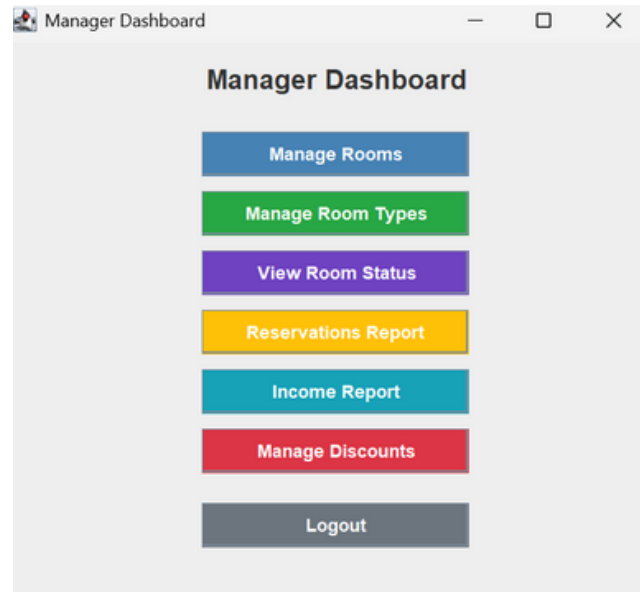
Overall, the GUI design emphasizes clarity, consistency, and usability, ensuring that hotel staff can interact with the system efficiently with minimal training.



Admin GUI

manager Dashboard (GUI Description)

The Manager Dashboard is the main control panel for hotel managers. It provides quick access to administrative and reporting functions required to manage hotel operations efficiently. Each button in the dashboard navigates to a specific module of the system, as described below:



Manage Rooms Screen (GUI Description)

The Manage Rooms screen allows the hotel manager to view and manage all rooms stored in the system database. The screen displays room information in a tabular format, providing a clear overview of room details and statuses.

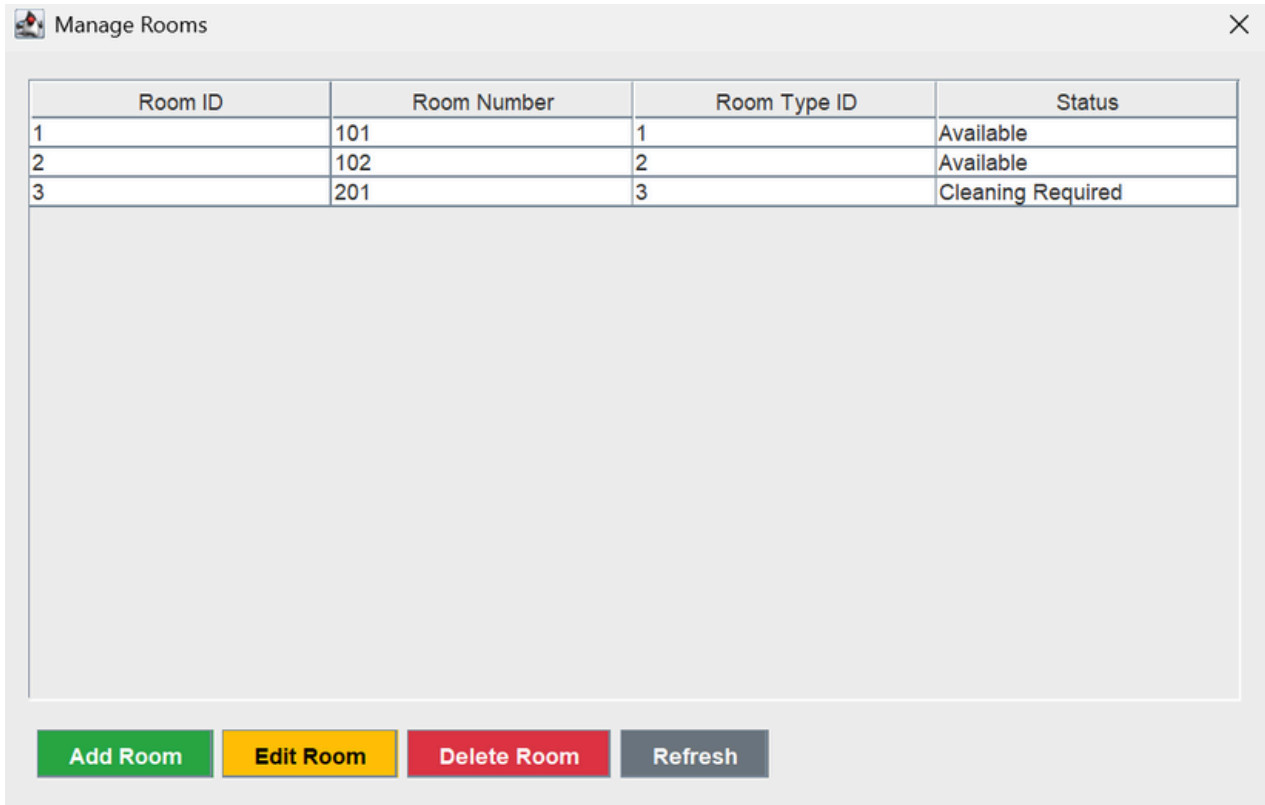
The table includes the following columns:

Room ID: A unique identifier for each room stored in the database.

Room Number: The actual room number used in the hotel.

Room Type ID: References the type/category of the room (e.g., single, double, suite).

Status: Indicates the current state of the room, such as Available, Occupied, or Cleaning Required.



Room ID	Room Number	Room Type ID	Status
1	101	1	Available
2	102	2	Available
3	201	3	Cleaning Required

Below the table, several action buttons are provided:

- Add Room** (Green button)
- Edit Room** (Yellow button)
- Delete Room** (Red button)
- Refresh** (Grey button)

Below the table, several action buttons are provided:

Add Room

This button opens a dialog that allows the manager to add a new room by entering the room number, room type ID, and initial status. The information is then saved to the database using the RoomDAO class.

Edit Room

This button allows the manager to modify the details of an existing room. The user must select a room from the table before editing. Changes are validated and then updated in the database.

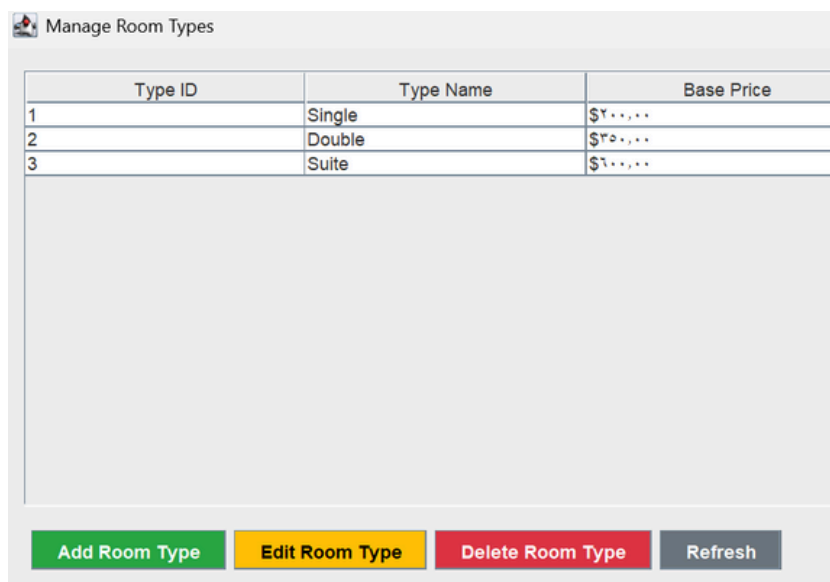
Delete Room

This option enables the manager to remove a selected room from the system. A confirmation dialog is displayed to prevent accidental deletion.

Refresh

This button reloads the room data from the database and updates the table to reflect the most recent changes.

This screen ensures accurate management of room data and plays a key role in maintaining correct room availability and status throughout the reservation lifecycle.



Type ID	Type Name	Base Price
1	Single	\$200.00
2	Double	\$300.00
3	Suite	\$100.00

Add Room Type
 Edit Room Type
 Delete Room Type
 Refresh

Manage Room Types Screen (GUI Description)

The Manage Room Types screen allows the hotel manager to define and manage different categories of rooms available in the hotel. This screen is essential for controlling room pricing and classification, which are later used during reservation creation and billing.

The room types are displayed in a table containing the following columns:

Type ID: A unique identifier for each room type stored in the database.

Type Name: The name of the room category, such as Single, Double, or Suite.

Base Price: The default price per night for the room type, which is used during invoice calculation.

At the bottom of the screen, several action buttons are available:

Add Room Type

This button allows the manager to create a new room type by specifying its name and base price. The new room type is saved in the database and becomes available for future reservations.

Edit Room Type

This option enables the manager to modify the details of an existing room type. The user must select a room type from the table before editing. Any updates to pricing or naming are reflected immediately in the system.

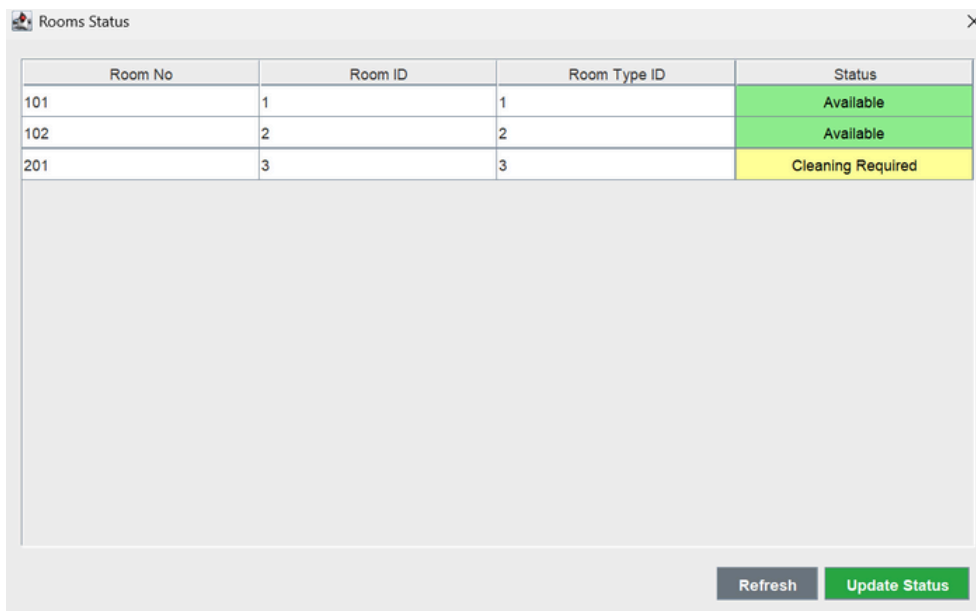
Delete Room Type

This button removes a selected room type from the system after confirmation. This prevents unused or obsolete room categories from remaining in the database.

Refresh

This button reloads the room type data from the database to ensure that the table displays the most up-to-date information.

This screen ensures flexible management of room categories and pricing, allowing the hotel to adjust offerings and prices efficiently.



Room No	Room ID	Room Type ID	Status
101	1	1	Available
102	2	2	Available
201	3	3	Cleaning Required

Refresh Update Status

Rooms Status Screen (GUI Description)

The Rooms Status screen provides the hotel staff and manager with a clear, real-time overview of the current status of all rooms in the hotel. This screen helps monitor room availability and operational readiness.

The screen displays a table containing the following columns:

- **Room No:** The physical room number used by hotel staff.
- **Room ID:** The unique internal identifier of the room stored in the database.
- **Room Type ID:** Identifies the type/category of the room.
- **Status:** Shows the current room status, such as Available, Occupied, or Cleaning Required.

To improve usability, the status column uses color coding:

- Green indicates rooms that are Available.
- Yellow indicates rooms that require cleaning.
- Other statuses (such as Occupied or Maintenance) are also visually distinguishable.

At the bottom of the screen, the following actions are available:

Refresh

Reloads the room data from the database to ensure that the displayed information is up to date.

Update Status

Allows the user to change the status of a selected room. This feature is typically used after check-in or check-out operations to update room availability or mark rooms as requiring cleaning.

This screen ensures accurate tracking of room usage and supports efficient hotel operations by providing instant visibility into room status changes.

Reservations Report (Analytics)

Report Range (YYYY-MM-DD)

From: 2025-12-08 To: 2025-12-14 Today This Week Refresh

Metric	Value
Report Range	2025-12-08 -- 2025-12-14
Total Rooms	3
Occupied Rooms (distinct overlap)	1
Occupancy Rate	33.33%
Total Room Nights (in range)	3
Room Revenue (from reservations.total_price)	\$18,000
ADR (Avg Daily Rate)	\$18,000
Invoices Count (created in range)	1
Invoices Total (sum invoices.total_amount)	\$16,000
Payments Received (sum payments.amount > 0)	\$16,000
Refunds (abs of payments.amount < 0)	\$0
Net Cash Flow (payments - refunds)	\$16,000

Notes

Occupancy: distinct rooms having reservations overlapping the selected range.
 ADR: total room revenue / total room nights within range.
 Revenue: invoices.total_amount and payments (positive=payments, negative=refunds).
 No-Show: approximated as reservations whose check-in date passed without CHECKED_IN/CHECKED_OUT/CANCELLED.

Export TXT Close

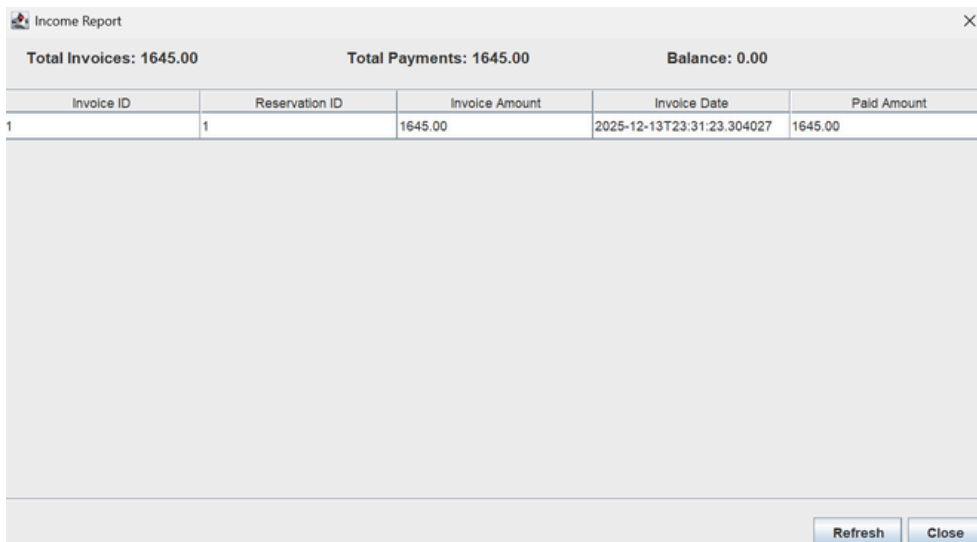
This screen provides analytical insights into hotel reservations and financial performance within a selected date range. The user can choose a custom period or quickly select Today or This Week, then refresh the report.

The report displays key metrics such as total rooms, occupied rooms, occupancy rate, total room nights, room revenue, average daily rate (ADR), number of invoices, total invoiced amounts, payments received, refunds, and net cash flow.

Calculated values are based on reservations, invoices, and payments stored in the database. A notes section explains important calculation rules, such as occupancy overlap and ADR formulas.

The report can be exported as a text file using the Export TXT button, and the Close button exits the screen.

This screen helps management monitor hotel performance and make informed operational and financial decisions.



The screenshot shows a window titled "Income Report" with a close button (X) in the top right corner. Below the title bar, there are three summary statistics: "Total Invoices: 1645.00", "Total Payments: 1645.00", and "Balance: 0.00". Below these statistics is a table with five columns: "Invoice ID", "Reservation ID", "Invoice Amount", "Invoice Date", and "Paid Amount". The table contains one row of data. Below the table, there are two buttons: "Refresh" and "Close".

Invoice ID	Reservation ID	Invoice Amount	Invoice Date	Paid Amount
1	1	1645.00	2025-12-13T23:31:23.304027	1645.00

The Income Report screen provides a financial overview of the hotel's revenue. It summarizes total invoice amounts, total payments received, and the remaining balance.

The table displays detailed invoice information, including invoice ID, reservation ID, invoice amount, invoice date, and the total paid amount for each invoice. All data is retrieved directly from the database to ensure accuracy.

The Refresh button reloads the latest financial data, while the Close button exits the report. This screen helps management monitor income and verify that payments match issued invoices.

ID	Code	Description	Percent	Active	Start Date	End Date
1	Rafeef	Opening Day	20.0	YES		

The Manage Discounts screen allows the manager to create, update, and manage discount offers applied to reservations and invoices.

The upper section contains input fields for entering discount details, including the discount code, percentage value, description, activation status, and validity period (start and end dates). Discounts can be enabled or disabled using the Active checkbox.

The table below displays all existing discounts stored in the database, showing their ID, code, description, percentage value, active status, and validity dates.

Available actions include:

Add: Creates a new discount record.

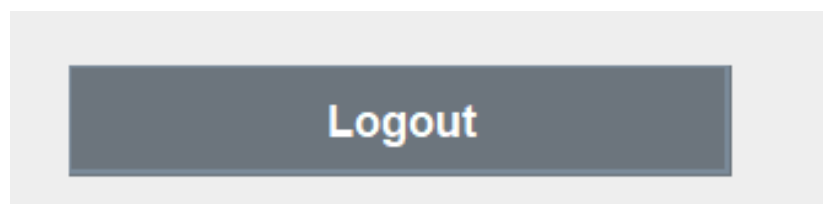
Update: Modifies the selected discount details.

Delete: Removes the selected discount from the system.

Clear: Clears all input fields.

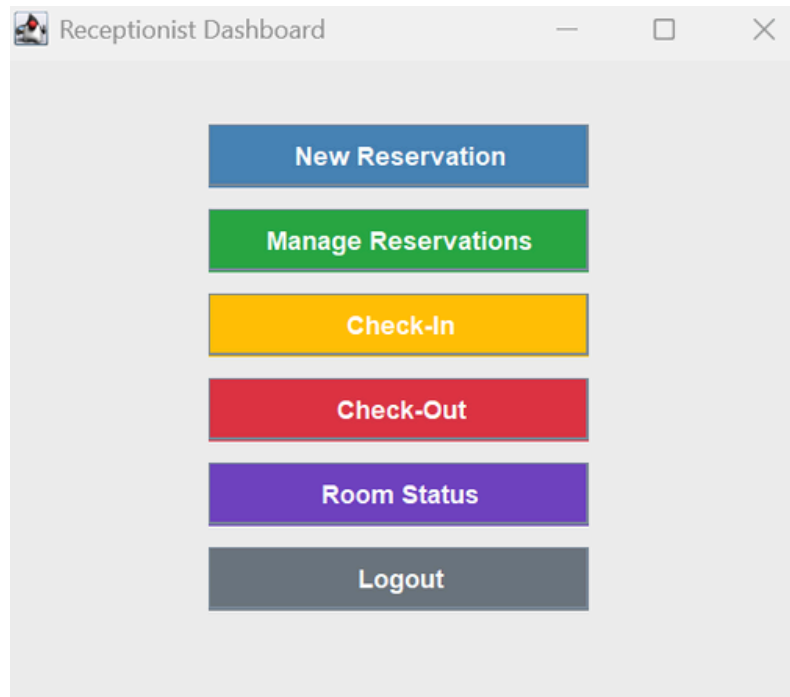
Close: Exits the screen.

This screen ensures flexible management of promotional discounts and supports accurate billing calculations during check-out.



The Logout button allows the current user to safely exit the system and return to the login screen. This ensures session security by preventing unauthorized access and allows another user to log in with different credentials.

Reception GUI



The Receptionist Dashboard provides quick access to all daily front-desk operations. It allows the receptionist to manage reservations and handle guest check-in and check-out processes efficiently.

Available actions include New Reservation, Manage Reservations, Check-In, Check-Out, Room Status, and Logout.

Each of these functions will be explained in more detail in the following sections.

This dashboard simplifies front-desk workflow and ensures smooth coordination between reservations and room operations.

New Reservation

Guest Information

Full Name:

Phone:

ID/Passport:

Email:

Room Selection

Room Type:

Check-in Date (YYYY-MM-DD):

Check-out Date (YYYY-MM-DD):

[Search Available Rooms](#) [View Waitlist](#)

Available Rooms

Room ID	Room No	Type	Price/Night
2	102	Double	350.0

[Cancel](#) [Confirm Reservation](#)

New Reservation Screen

The New Reservation screen is used to create a new hotel reservation. It allows staff to enter guest information such as full name, phone number, ID/passport, and email.

The user selects the room type and specifies the check-in and check-out dates. By clicking Search Available Rooms, the system checks room availability based on the selected dates and displays matching rooms with their price per night.

Once an available room is selected, the reservation can be completed using the Confirm Reservation button. The View Waitlist button allows staff to review waiting reservations if no rooms are available, while Cancel exits the screen without saving.

This screen ensures accurate reservation creation and real-time room availability tracking through database integration.

The screenshot shows a web application window titled "Manage Reservations". At the top, there is a "Search Reservations" section with a text input field labeled "Search:" and a blue "Search" button. Below this is a table with the following data:

Reservation ID	Guest Name	Room No	Check-in	Check-out	Status	Total Price
1	Kadi	201	2025-12-12	2025-12-15	CHECKED_OUT	1800.0

Below the table, there is a large empty rectangular area. At the bottom of the window, there is a row of action buttons: "View Details", "Edit", "Cancel Reservation" (highlighted in yellow), "Go to Check-In" (highlighted in green), "Go to Check-Out" (highlighted in blue), and "View Waitlist".

Manage Reservations Screen

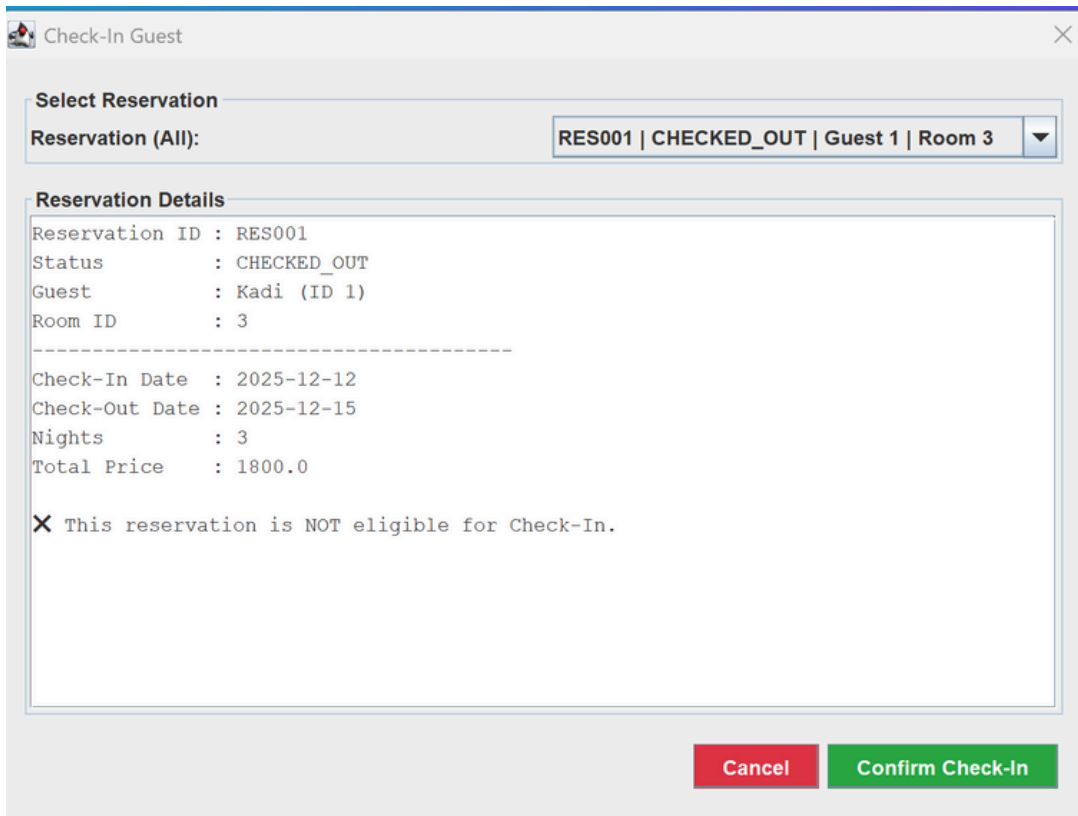
The Manage Reservations screen allows staff to search, view, and manage existing reservations. Reservations can be searched using keywords such as guest name or reservation ID.

The table displays key reservation details, including reservation ID, guest name, room number, check-in and check-out dates, status, and total price.

Available actions include:

- **View Details:** Displays full reservation information.
- **Edit:** Updates reservation details when allowed.
- **Cancel Reservation:** Cancels a reservation if it has not been checked in.
- **Go to Check-In:** Proceeds with guest check-in for eligible reservations.
- **Go to Check-Out:** Initiates the check-out process for checked-in reservations.
- **View Waitlist:** Displays reservations placed on the waiting list.

This screen provides full control over the reservation lifecycle and ensures smooth front-desk operations.



Check-In Guest

Select Reservation

Reservation (All): RES001 | CHECKED_OUT | Guest 1 | Room 3

Reservation Details

Reservation ID : RES001
Status : CHECKED_OUT
Guest : Kadi (ID 1)
Room ID : 3

Check-In Date : 2025-12-12
Check-Out Date : 2025-12-15
Nights : 3
Total Price : 1800.0

X This reservation is NOT eligible for Check-In.

Cancel Confirm Check-In

The Check-In Guest screen is used to register guest arrivals based on existing reservations. The user selects a reservation from the list, and the system displays full reservation details including guest information, room number, stay dates, number of nights, total price, and current reservation status.

The system performs status validation before allowing check-in. If the reservation is not eligible (for example, already checked out or canceled), a clear message is displayed to prevent invalid operations. The Confirm Check-In button completes the process for eligible reservations and updates the room status to Occupied, while Cancel exits the screen without changes.

This screen ensures controlled and accurate check-in operations aligned with reservation status rules.

Check-Out

Select Reservation

Reservation (CHECKED_IN): 2 | Guest 2 | Room 2

Nights: 3 Room Price/Night: \$20.00

Charges Breakdown (Itemized)

Description	Amount
Room Charges	\$10.00
Extra Services	\$0.00
Tax (10%)	\$1.00
Service Fee	\$2.00
SUBTOTAL	\$13.00
TOTAL AMOUNT	\$13.00

Discount Code: Late Penalty: 0.00

Extra Services (example: Laundry=30)

Payment Summary

Total: \$13.00 Paid: \$0.00

Refunds: \$0.00 Balance: \$13.00

Generate Invoice Pay Refund Complete Check-Out Cancel

The Check-Out screen is used to complete the guest departure process. The user selects a reservation with status CHECKED_IN, and the system automatically loads stay details such as number of nights and room price per night.

An itemized charges table displays all billing components, including room charges, extra services, taxes, penalties, and applied discounts. The user can enter a discount code, add extra services, or specify a late check-out penalty.

A payment summary section shows the total amount, paid amount, refunds, and remaining balance in real time.

Available actions include:

Generate Invoice: Creates an invoice for the reservation.

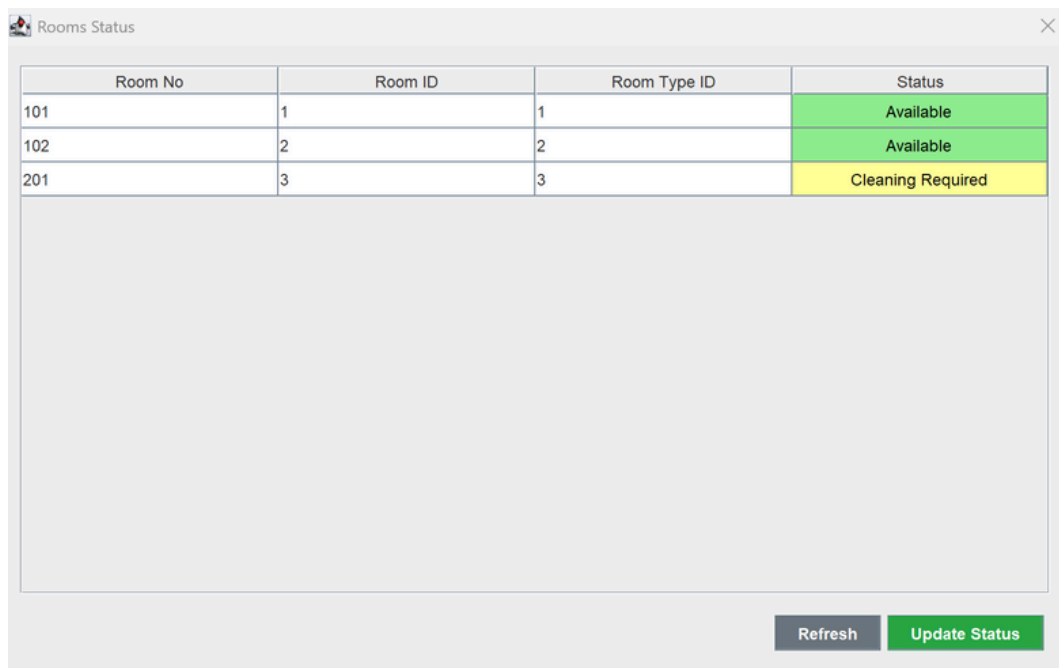
Pay: Records a payment toward the invoice.

Refund: Processes a refund if applicable.

Complete Check-Out: Finalizes the check-out and updates room status to Cleaning Required.

Cancel: Exits without changes.

This screen ensures accurate billing, controlled payments, and proper room status updates during the check-out process.



Room No	Room ID	Room Type ID	Status
101	1	1	Available
102	2	2	Available
201	3	3	Cleaning Required

Refresh Update Status

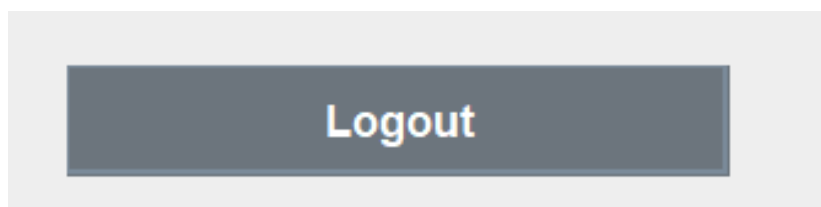
Rooms Status Screen

The Rooms Status screen displays the current status of all hotel rooms in real time. It shows the room number, room ID, room type, and current status.

Room statuses are visually highlighted using colors to improve clarity, such as green for Available rooms and yellow for Cleaning Required rooms.

The Refresh button reloads the latest data from the database, while Update Status allows staff to change the status of a selected room after check-in or check-out.

This screen helps ensure accurate tracking of room availability and operational readiness.



The Logout button allows the current user to safely exit the system and return to the login screen. This ensures session security by preventing unauthorized access and allows another user to log in with different credentials.

Database Connection

The system connects to the database using SQLite JDBC. A dedicated class named `DatabaseConnection` was implemented to manage database connectivity across the application.

The connection is established using the `DriverManager` class, which opens a connection to the SQLite database file located in the project resources directory (`hotel.db`). The database URL follows the SQLite JDBC format:

```
jdbc:sqlite:src/main/resources/hotel.db
```

The `getConnection()` method provides a reusable way for Data Access Object (DAO) classes to access the database. Each DAO class (such as `ReservationDAO`, `RoomDAO`, and `GuestDAO`) is responsible for executing SQL operations related to a specific entity, including retrieving, inserting, updating, and deleting records.

This design separates database logic from the user interface, centralizes database operations, and improves maintainability and code organization.

To verify and inspect the database content, DB Browser for SQLite was used. This tool helped ensure that data was stored correctly and that database operations were executed as expected.



Exception Handling

The Hotel Reservation System implements structured exception handling to ensure application stability, prevent unexpected crashes, and provide meaningful feedback to users. The system uses try-catch blocks extensively across the application, particularly in database operations and user input processing.

Several runtime exceptions may occur during system execution due to user input errors or database operations. The system handles these exceptions to prevent crashes and ensure data integrity.

- **Invalid numeric input:**

When the user enters non-numeric values in fields such as payment amount or deposit, a `NumberFormatException` may occur. The system handles this exception by displaying a validation message and preventing the operation from proceeding.

- **Missing required fields:**

If mandatory fields (such as guest ID or reservation selection) are left empty, the system prevents the action and prompts the user to complete the required information.

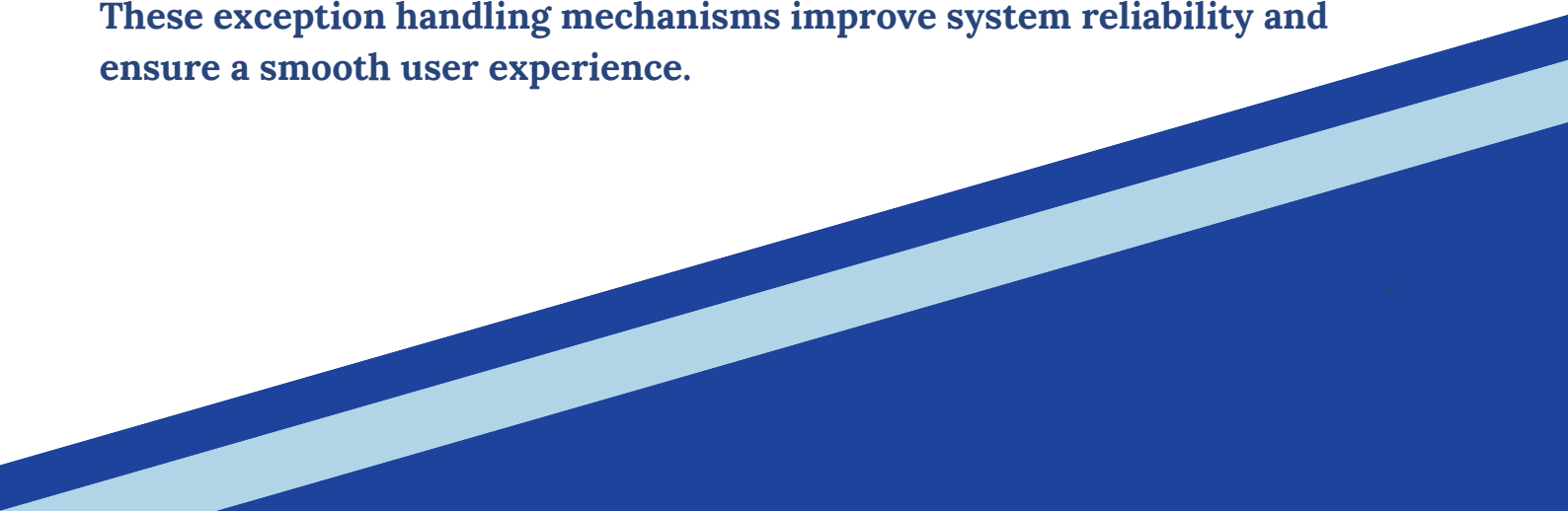
- **Database operation errors:**

During database interactions such as inserting invoices or payments, a `SQLException` may occur due to connection issues or invalid data. These exceptions are caught to prevent system termination, and an error message is displayed to the user.

- **Invalid reservation or invoice selection:**

If an operation is attempted on a non-existing reservation or invoice, the system safely handles the situation by notifying the user and stopping the process.

These exception handling mechanisms improve system reliability and ensure a smooth user experience.



Exception Handling

Database-related exceptions, such as connection failures or SQL execution errors, are caught and handled gracefully. When an error occurs, the system displays clear error messages using dialog boxes, allowing users to understand the issue without exposing technical details.

User input validation is performed before executing critical operations to prevent common runtime errors. This includes checking for empty fields, invalid numeric values, incorrect date formats, and invalid discount codes. If validation fails, informative warning messages are shown to guide the user in correcting the input.

Transactional operations, such as check-out and invoice generation, are protected using database transactions. In case of failure, changes are rolled back to maintain data consistency between reservations, rooms, invoices, and payments.

Through these mechanisms, the system ensures reliability, data integrity, and a smooth user experience even when errors occur.



Testing and Validation

Testing was performed to verify that the Hotel Reservation System operates correctly and meets functional requirements. Multiple testing approaches were applied, including unit testing, integration testing, and validation testing.

Reservation and Room Booking Testing

- The system allows room reservations only when rooms are available for the selected dates.
- Date overlap validation prevents double-booking of the same room.
- Reservations are created with correct guest information, room details, check-in and check-out dates.
- Reservation status transitions (CONFIRMED, CHECKED_IN, CHECKED_OUT, CANCELLED) work correctly.
- Room status updates automatically based on reservation actions (Available → Occupied → Cleaning Required).

Billing and Payments Testing

- Room charges are calculated based on the number of nights and room price.
- Discounts are applied correctly when valid discount codes are used.
- Invoice totals are accurate and stored correctly.
- Payments update invoice balances accurately and support multiple payments.

Validation and Error Handling

- Invalid reservation dates are rejected.
- Required fields must be filled before saving reservations.
- Numeric values (prices, discounts, payments) are validated.
- Confirmation dialogs prevent accidental actions.

Conclusion

The testing results confirm that room booking, billing, and payment processes function correctly. The system effectively prevents booking conflicts and ensures accurate financial calculations, making it suitable for real hotel operations.



Conclusion

In conclusion, the Hotel Reservation and Check-In System successfully demonstrates the application of core software engineering principles in designing and implementing a real-world hotel management solution. The system provides essential functionalities such as room reservations, guest management, check-in and check-out operations, waitlist handling, and billing support, ensuring an organized and efficient workflow for hotel staff.

By integrating a relational database, the system ensures data consistency, persistence, and reliability across all operations. The use of a layered architecture with clearly separated models, data access objects (DAOs), and user interface components enhances maintainability and scalability. Additionally, validation mechanisms and exception handling improve system robustness by preventing invalid bookings and handling conflicting reservation scenarios.

Overall, this project reflects a practical understanding of object-oriented design, database integration, and GUI development using Java Swing. It effectively simulates real-world hotel operations and serves as a strong foundation for future enhancements such as online reservations, payment gateway integration, and advanced reporting features.

References

- Course lecture slides prepared by Dr. Walid Karamti and Dr. Hedia Zardi
- Stack overflow